

QIEMO QARQAN, China

WMO: 518550

Lat: 38.1329N

Lon: 85.5372E

Elev: 4094

StdP: 12.65

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	6.5	-12.1	3.3	24.7	-8.7	4.0	22.6	15.7	27.5	13.2	28.6	0.6	N/A	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	95.4	61.9	93.4	61.5	91.0	60.9	66.2	86.9	64.7	85.4	63.4	83.9	6.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	90.5	73.0	57.8	83.3	72.0	55.8	77.1	72.3	33.3	86.7	32.1	85.8	31.0	83.7	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	15.8	12.8	-1.5	101.3	4.0	2.1	-4.4	102.8	-6.7	104.1	-8.9	105.3	-11.8	106.9
			-2.7	71.0	3.4	3.1	-5.2	73.2	-7.2	75.1	-9.1	76.8	-11.6	79.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	19.9	28.4	42.4	56.6	66.9	73.0	76.5	74.7	64.8	50.4	36.3	23.2
	DBStd	20.68	5.41	5.64	7.56	6.70	5.90	4.76	3.89	4.33	4.87	6.72	7.32	6.27
	HDD50	3117	933	604	248	13	1	0	0	0	0	74	414	830
	HDD65	6114	1398	1024	701	265	47	4	0	2	62	454	862	1295
	CDD50	3557	0	0	13	211	524	690	821	765	444	87	2	0
	CDD65	1076	0	0	0	12	105	244	357	301	56	1	0	0
	CDH74	14265	0	0	0	340	1525	2907	4347	4024	1036	86	0	0
	CDH80	6293	0	0	0	107	534	1246	2140	1974	276	16	0	0
Wind	WSAvg	3.9	2.8	3.7	4.9	5.3	4.9	4.1	4.1	4.3	4.0	3.5	2.6	2.2
Precipitation	PrecAvg	0.9	0.0	0.0	0.0	0.0	0.1	0.2	0.3	0.2	0.0	0.0	0.0	0.0
	PrecMax	1.9	0.2	0.1	0.3	0.7	0.7	0.7	1.2	0.6	0.2	0.1	0.2	0.1
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.5	0.0	0.0	0.1	0.2	0.1	0.2	0.3	0.2	0.0	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	52.9	70.0	89.6	92.7	97.8	98.7	98.4	89.9	81.0	65.3	46.7
		MCWB	29.7	36.7	45.1	55.8	57.6	64.4	64.6	62.9	59.2	52.4	44.5	33.7
	2%	DB	37.6	47.8	65.9	83.1	89.0	93.1	95.7	95.2	86.1	76.0	58.6	40.7
		MCWB	27.4	33.4	43.2	52.7	57.4	62.0	62.7	61.3	57.4	50.2	41.1	30.1
	5%	DB	34.2	44.5	61.8	77.4	85.9	90.2	93.4	93.0	83.2	71.5	53.8	37.2
		MCWB	25.5	31.6	41.9	50.2	56.8	60.9	62.3	60.8	55.8	48.5	39.0	28.3
	10%	DB	31.3	41.5	58.3	73.3	82.6	87.2	90.9	90.4	80.3	66.7	49.7	34.2
		MCWB	23.7	29.8	39.9	48.0	55.2	59.7	62.2	60.2	54.8	46.4	37.0	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.2	37.2	45.7	55.7	62.0	67.3	69.3	67.0	62.1	54.4	44.8	34.8
		MCDB	41.2	51.0	68.1	86.3	86.6	89.5	87.1	88.2	85.4	78.6	64.4	45.1
	2%	WB	27.7	34.1	43.8	53.0	59.2	64.6	67.0	64.8	59.2	50.9	41.4	31.0
		MCDB	37.1	46.9	64.3	80.8	84.9	87.2	87.0	87.1	82.2	73.7	56.9	39.3
	5%	WB	25.7	31.9	42.2	50.9	57.5	62.9	65.4	63.2	57.3	48.7	39.4	28.8
		MCDB	34.0	43.7	61.4	75.9	82.3	85.2	85.5	86.0	78.6	69.9	52.9	37.0
	10%	WB	23.8	30.0	40.1	48.8	55.9	61.4	64.2	61.9	55.7	46.9	37.2	26.8
		MCDB	31.1	41.0	57.3	71.2	79.4	82.6	83.6	84.7	76.5	65.7	49.3	34.2
Mean Daily Temperature Range	5% DB	MDBR	21.6	23.1	24.8	26.7	25.3	24.2	24.6	27.4	29.2	28.4	23.8	20.1
		MCDBR	25.1	27.3	30.0	33.8	32.2	30.4	31.2	33.4	33.6	33.6	29.6	24.4
		MCWBR	17.7	16.7	15.0	14.3	13.0	11.1	9.5	11.0	12.7	15.3	16.5	16.2
	5% WB	MCDBR	24.9	26.0	28.5	32.0	27.6	24.3	23.3	26.8	29.5	32.8	28.9	23.8
		MCWBR	17.6	16.1	14.4	13.8	11.5	10.7	9.2	10.2	12.4	15.1	16.2	15.8
Clear-Sky Solar Irradiance	taub	0.343	0.407	0.505	0.600	0.628	0.646	0.628	0.613	0.568	0.455	0.378	0.350	
	taud	2.060	1.898	1.677	1.511	1.513	1.529	1.592	1.602	1.649	1.817	1.984	2.039	
	Ebn at Noon	255	250	236	223	220	216	219	217	216	230	237	240	
	Edh at Noon	38	52	71	89	91	89	83	81	73	55	41	36	
All-Sky Solar Radiation	RadAvg	891	1192	1639	1962	2173	2193	2137	1978	1794	1471	1042	817	
	RadStd	95	63	75	98	101	104	117	127	61	36	49	50	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	-1.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-250	-211

Nomenclature: See separate page